Heritage Bank Digital — Customer & Operator Journey Map

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Overview

This document maps the primary journeys for retail and SMB customers, and the branch teller who supports them, across Heritage Bank's modernised digital channels. All journeys route through the Banking Integration Hub (BIH) to the IBM AS/400 core; the AS/400 remains the system of record throughout.

Journey 1 — Retail Customer: Check Balance and Transfer Funds

Persona: Carol, 58, long-standing Heritage customer; not tech-savvy but comfortable with mobile banking

Goal: Check her savings balance and move $500 to her checking account before a bill payment clears.

Step	Action	Service	Detail
1	Opens Heritage mobile app; FaceID login	Identity Service (Okta + biometric)	OIDC PKCE flow; biometric as 2nd factor
2	Dashboard loads: all account summaries	Account Service → BIH → AS/400	Available balance shown (not book balance)
3	Taps "Move Money" → selects savings-to-checking	Payment Service	—
4	Enters $500; reviews and confirms	Payment Service	—
5	Transfer posts immediately; balance updates	Payment Service → BIH → FiServ API	AS/400 books transaction; available balance refreshed
6	Push notification: "Transfer of $500 complete"	Notification Service	Within 10 seconds
7	Carol checks transaction history; sees transfer	Account Service	Transaction visible immediately (no EOD wait)

Business rules in scope:

- Available balance updated in real time via FiServ API; book balance not shown to customers
- If transfer attempted during AS/400 batch window (00:00–04:00 CT), Payment Service queues it and shows "Scheduled — will process after maintenance window"
- Overdraft protection: if checking balance would go negative, system checks linked savings first; auto-transfer covers shortfall (max 3 auto-transfers/day)

Journey 2 — Retail Customer: Initiate an ACH External Transfer

Persona: Tom, 34, freelancer; moves money between Heritage and his business account at another bank

Goal: Transfer $3,000 from his Heritage checking to an external business account for next-day ACH.

Step	Action	Service	Detail
1	Navigates to "Transfers → External" in online banking	Payment Service	External account already linked
2	Selects linked external account; enters $3,000	Payment Service	Velocity check: ACH credit limit $25,000/day
3	Selects "Next Business Day" delivery	Payment Service	SDIE (same-day) option shown only if before 2:30 PM CT
4	Confirms transfer	Payment Service	Idempotency key generated; stored in Redis
5	ACH file built; submitted to AS/400 ACH module via IBM MQ	BIH → AS/400	AS/400 submits to FedACH at scheduled window
6	Confirmation email sent; ACH tracking ID shown	Notification Service	—
7	Next morning: funds debited from Heritage checking	AS/400 (nightly batch)	Book balance updated; transaction posted
8	ACH return (if any) triggers notification within 24h	Payment Service → Notification Service	—

Key acceptance criteria:

- SDIE option must be disabled in the UI after 2:30 PM CT (not just server-side — prevent confusion)
- Velocity limit (ACH credit $25,000/day, ACH debit $10,000/day) checked at Payment Service; clear error message with current day's total and remaining limit
- Duplicate submission protection: same amount + same external account + same day → block with "duplicate transfer" warning
- ACH return codes must be translated to plain-English messages (e.g., R01 "Insufficient funds at receiving bank")

Journey 3 — Fraud Alert and Card Block

Persona: David, 41, whose debit card was used fraudulently while he was travelling

Goal: Lock his card immediately when he receives a fraud alert; dispute the transaction.

Step	Action	Service	Detail

1	NICE Actimize detects anomalous transaction ($890 at foreign merchant)	Fraud & Compliance Service	Actimize pushes alert via REST to Fraud Service
2	Heritage push notification + SMS: "Unusual activity detected"	Notification Service	Within 60 seconds of Actimize alert
3	David opens app; sees fraud alert banner	Card Management Service	—
4	Taps "Lock Card"	Card Management Service → FIS Worldpay	Card status = BLOCKED in Worldpay within 5 seconds
5	Taps "Dispute Transaction"	Payment Service	Reg E dispute opened; provisional credit initiated
6	CSA notified; case created in Salesforce	Fraud & Compliance Service → Salesforce	Linked to transaction ID
7	Provisional credit posted within 10 business days	Account Service → BIH	Reg E requirement
8	Investigation completed; permanent credit or debit	Compliance team action	Written resolution sent to David

Acceptance criteria:

- Card lock must take effect at FIS Worldpay within 5 seconds of user tap; subsequent transactions declined immediately
- Fraud alert notification must arrive within 60 seconds of Actimize alert ingestion on Kafka
- Reg E dispute form must capture: transaction date, amount, merchant, description of dispute; auto-populate from transaction record
- Provisional credit must not exceed the disputed transaction amount; shown as pending in account history with label "Dispute — provisional credit"

Journey 4 — SMB Owner: Set Up an Authorised User

Persona: Rachel, 44, owner of a 12-person landscaping business; uses Heritage Business Banking

Goal: Give her office manager Maria view-only access to check balances, and her bookkeeper James initiator access for ACH payroll — but not approver access.

Step	Action	Service	Detail
1	Rachel logs into SMB Portal	Identity Service	Cognito business user pool; MFA required
2	Navigates to "Team Access" → "Add User"	Customer Service	—
3	Enters Maria's email; selects "Viewer" role	Customer Service → SMB Portal BFF	Viewer = read-only; no transaction initiation

4	Maria receives invite; completes registration with MFA	Identity Service	—
5	Enters James's email; selects "Initiator" role	Customer Service	Initiator = create transfers, cannot approve
6	James cannot be assigned "Approver" at the same time	SMB Portal BFF	Business rule enforced at API layer, not just UI
7	Rachel as Owner approves James's ACH payroll run	Payment Service	Dual auth: Initiator creates, Owner/Approver approves
8	ACH payroll file uploaded by James (NACHA format)	Payment Service → BIH → AS/400 ACH	≤ $25,000 approval by Rachel; above $25,000 requires Wire Desk

Acceptance criteria:

- Role assignment (Viewer / Initiator / Approver) enforced at Payment Service API — UI enforcement alone is not sufficient
- A user with Initiator role must not see the "Approve" button on any transaction they created
- Removing a user must immediately revoke all active sessions for that user (Okta token revocation)
- NACHA file validation: ACH Service must parse and validate file format before submitting to BIH; return descriptive errors for malformed entries

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Journey 5 — Branch Teller: Assisted Service

Persona: Sandra, branch teller at Heritage's Columbus flagship branch

Goal: Help a customer who forgot their mobile login and wants to check their balance and request a wire.

Step	Action	Service	Detail
1	Sandra authenticates to Branch Teller App (mTLS + SAML SSO via Active Directory)	Identity Service	Internal only; no customer-facing Okta
2	Searches customer by name + last 4 of SSN	Customer Service → BIH → AS/400	Read-only identity lookup
3	Verifies customer identity (government ID + verbal KBA)	Customer Service	ID document scan logged as Audit event
4	Reads account balances and recent transactions on behalf of customer	Account Service	Audit event: actor = Sandra's employee ID, patient = customer MRN/account
5	Customer requests $15,000 wire to vendor	Payment Service	Wire above $10,000 requires Branch Manager PIN approval
6	Sandra initiates wire; Branch Manager Janet approves with PIN	Payment Service	Dual-auth enforced by Payment Service; branch manager role checked

7	Wire submitted to Wire Desk; confirmation number issued	Payment Service → BIH → AS/400 Fedwire	Wire Desk processes same day if before 14:30 CT
8	Sandra ends session; 8-minute inactivity timeout	Identity Service	Stricter timeout for internal teller sessions

Acceptance criteria:

- Teller session must timeout after 8 minutes inactivity (stricter than customer 10-minute rule)
- All teller-on-behalf-of-customer actions must log both the teller ID and the customer account number to Audit Service
- Branch Manager approval must be captured as a separate authentication event (not just a PIN field in a form)
- Wire confirmation number from AS/400 must be displayed to Sandra and emailed to the customer automatically

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Legacy Integration Pain Points (Current State)

The following table documents pain points in the current AS/400-connected system that the BIH modernisation aims to resolve:

Pain point	Current behaviour	Target behaviour after BIH
Balance displayed to customers	Book balance (as of EOD yesterday)	Available balance (real-time via FiServ API)
Transfer during batch window	Teller sees "Host Unavailable" error; customer must call back	Queue + retry; customer sees "Scheduled for after maintenance"
ACH return codes	Raw NACHA R-codes shown in teller system	Plain-English translation in both teller app and customer app
Card lock	Teller calls FIS Worldpay support line; 10-minute wait	Card Management Service API to Worldpay < 5 seconds
Wire dual auth	Manager walks to teller terminal and types on same screen	Separate approval from manager's own device
Audit trail	AS/400 produces SMF log; reviewed weekly by compliance	Real-time Kafka stream → Audit Service → Elasticsearch